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(54) **LATENCY COMPENSATION BETWEEN
COORDINATED PLAYER DEVICES**

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H04N 21/43 (2011.01)
H04R 3/00 (2006.01)
H04R 29/00 (2006.01)

(52) **U.S. Cl.**

CPC **H04N 21/4307** (2013.01); **G11B 27/10** (2013.01); **H04N 21/41407** (2013.01); **H04R 3/00** (2013.01); **H04R 29/001** (2013.01)

(58) **Field of Classification Search**

CPC H04N 21/4307; H04N 21/41407; G11B 27/10; H04R 3/00; H04R 29/001

See application file for complete search history.

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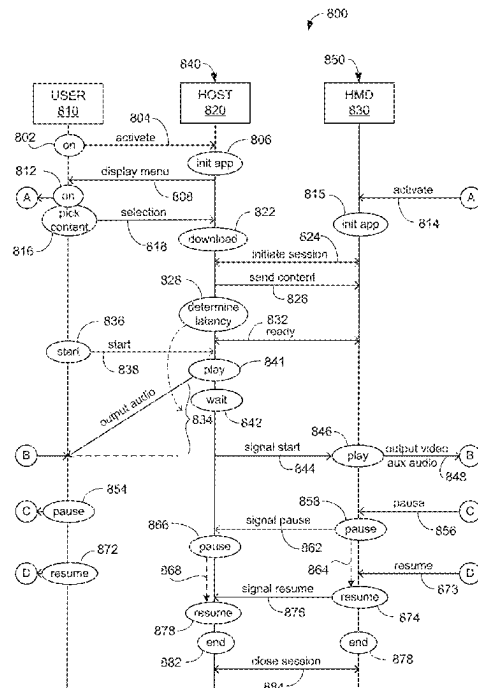
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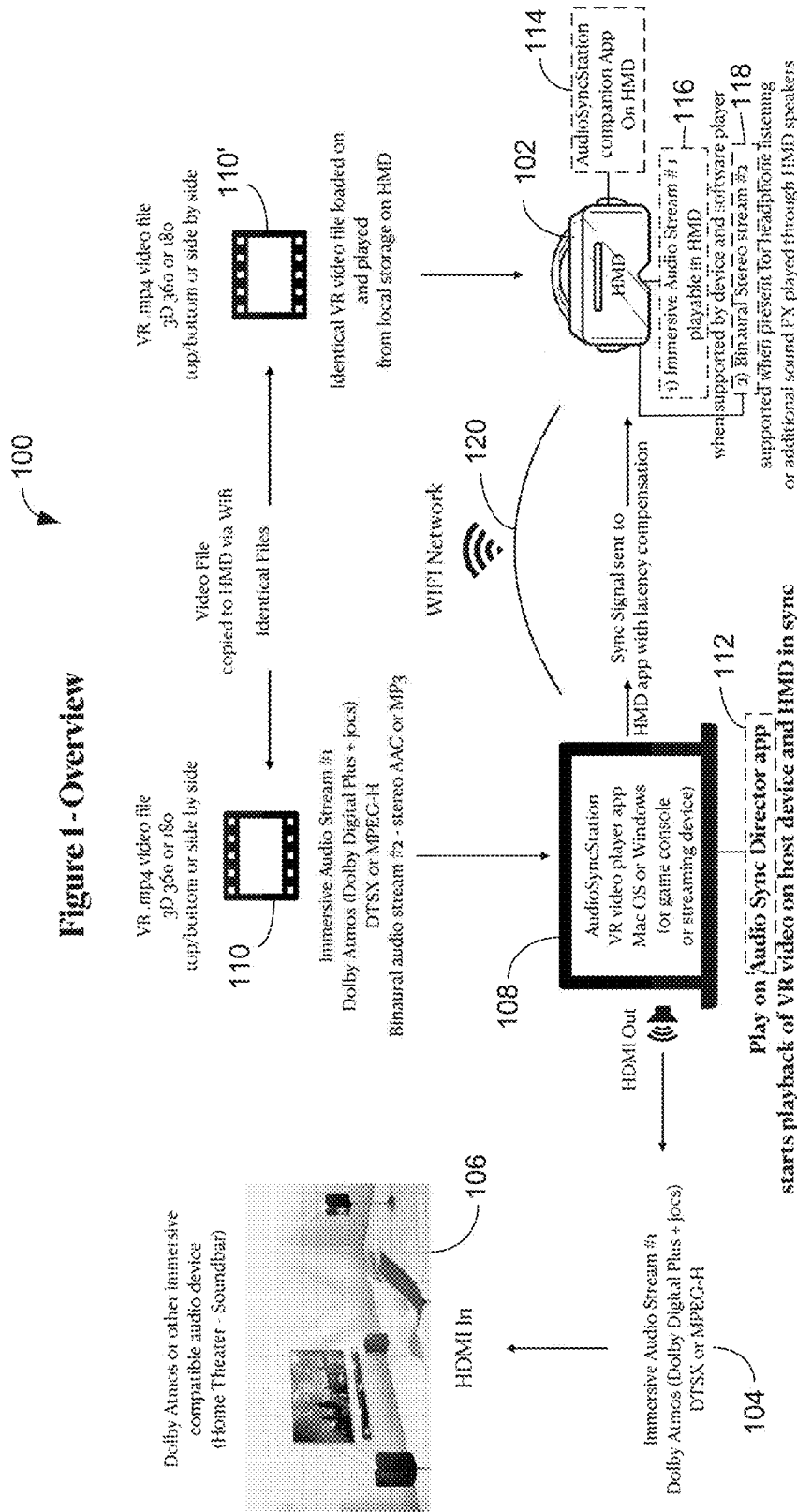
(57)

ABSTRACT

An apparatus and method for simultaneous output of video from a head-mounted display device and audio output from an external sound system includes compensating for latency in the external sound system relative to the head-mounted display. The latency is measured by outputting test content using one of alternative methods.

20 Claims, 12 Drawing Sheets





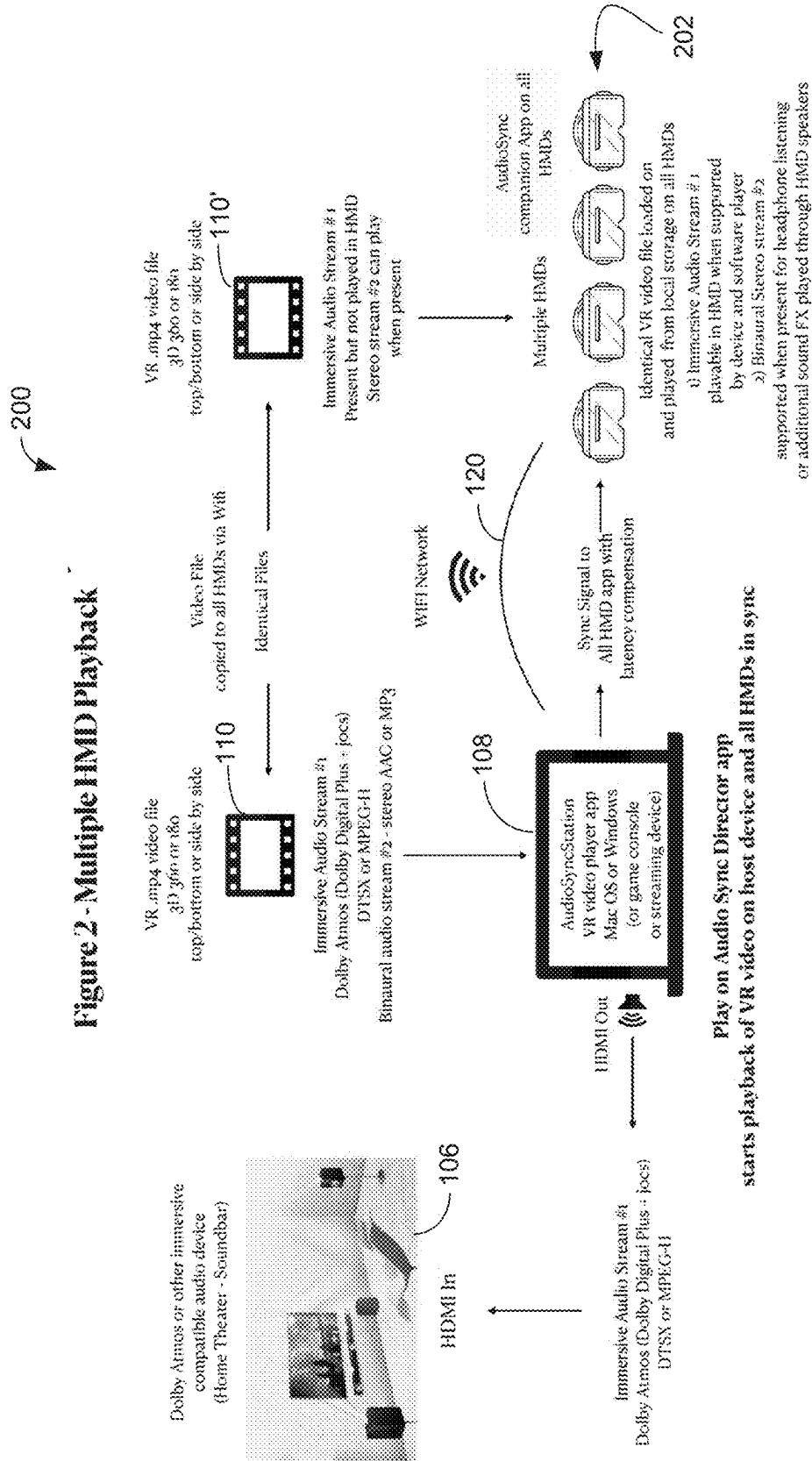


Figure 5 - VR Media File Detail & Workflow

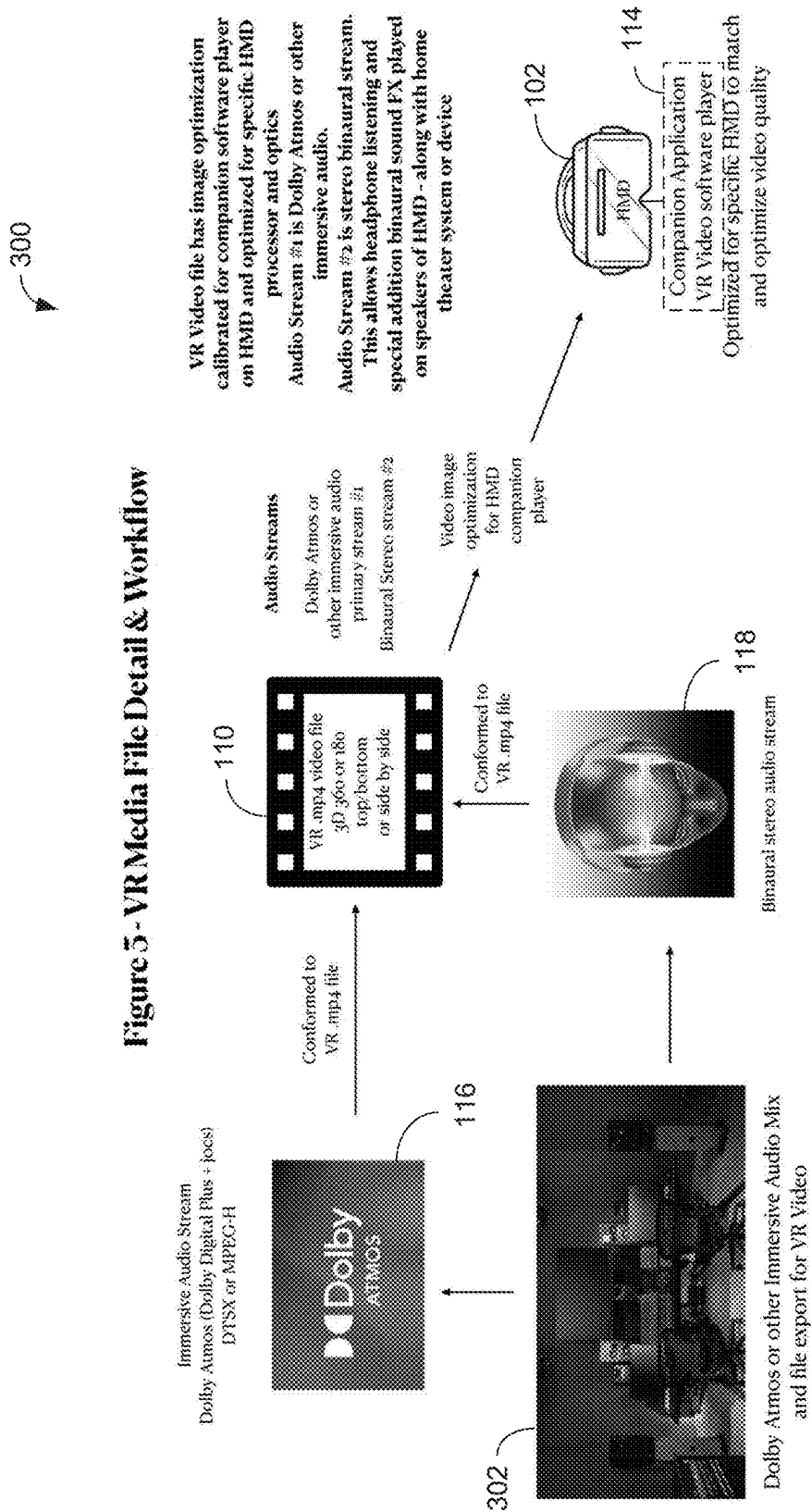


FIG. 4 - Immersive Audio Speaker and Binaural Hybrid

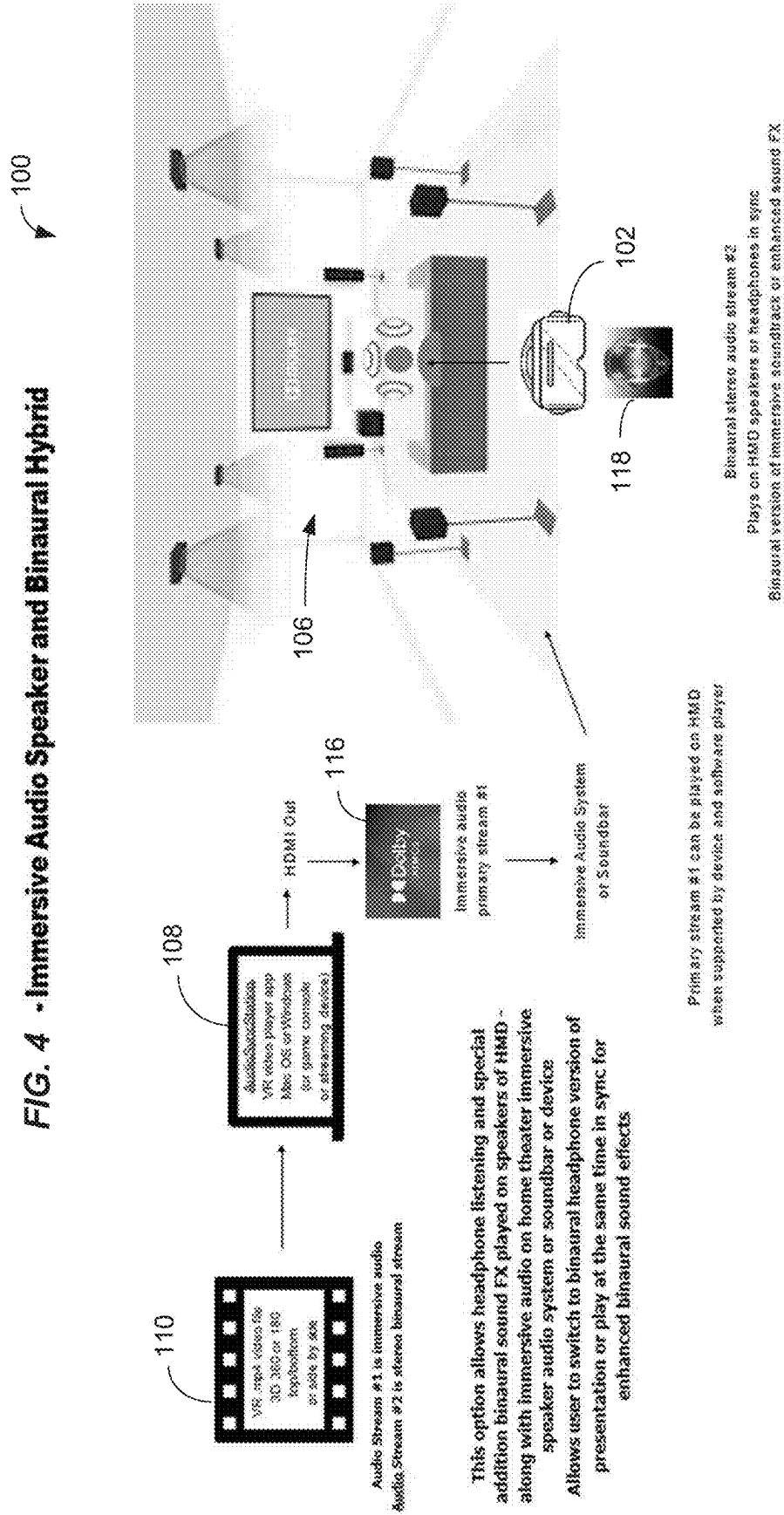


FIG. 6 -Media Loading and Sync

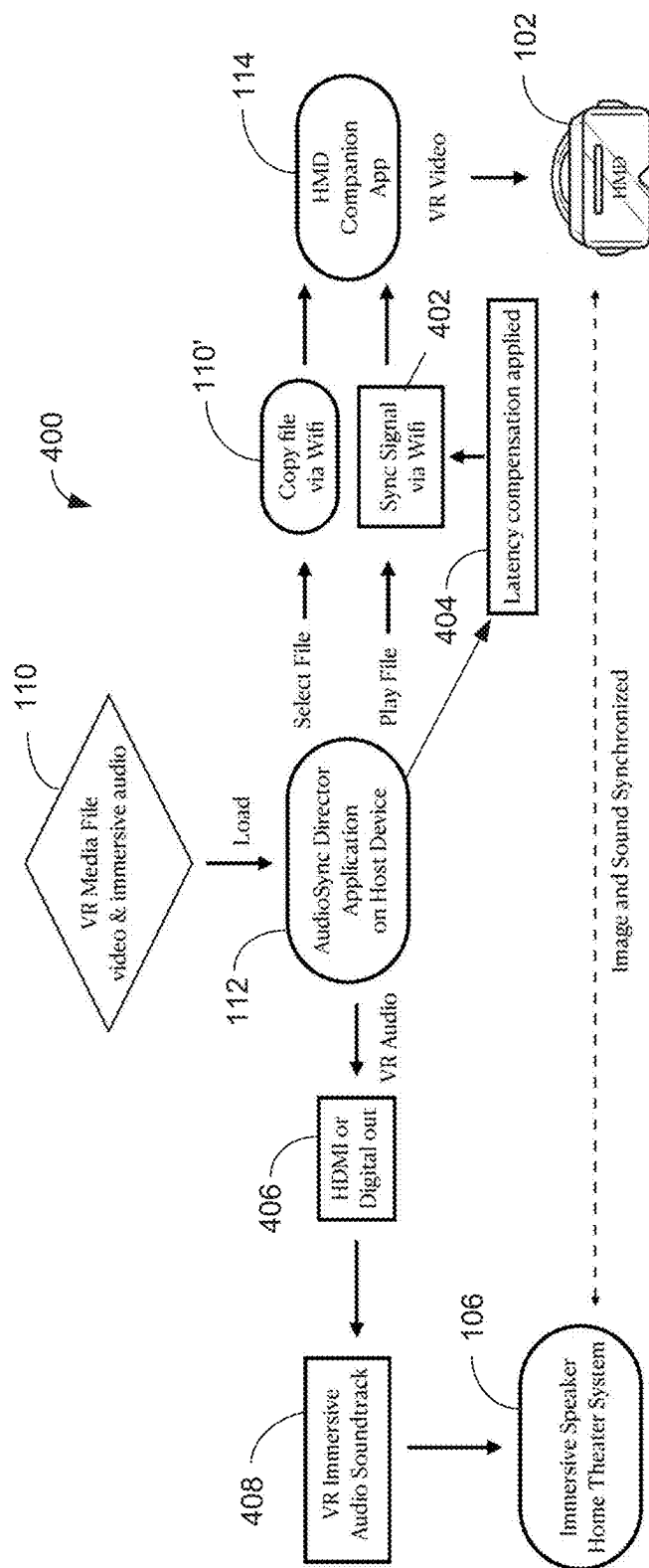


FIG. 7 - Media Management and Playback

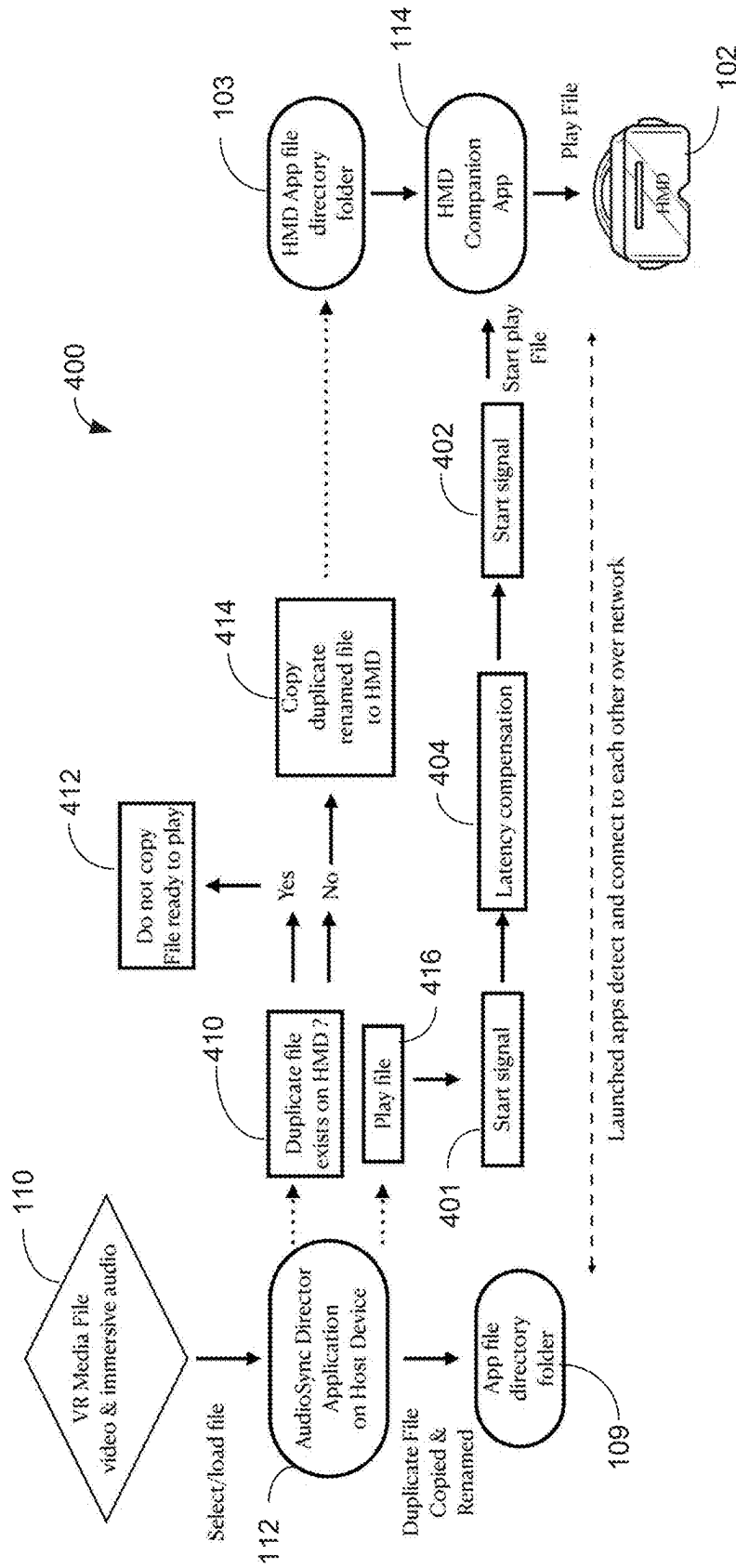


FIG. 8

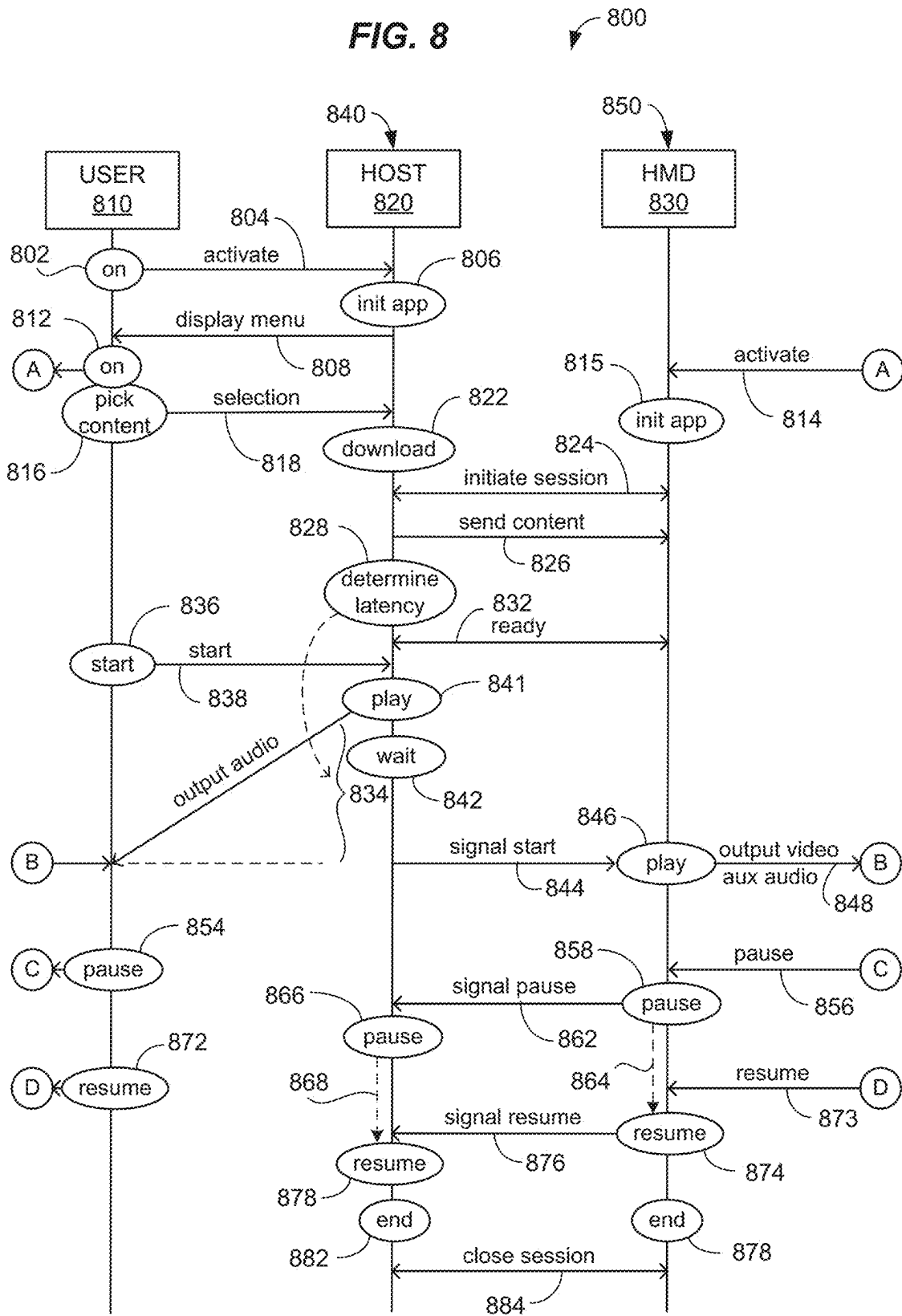


FIG. 9

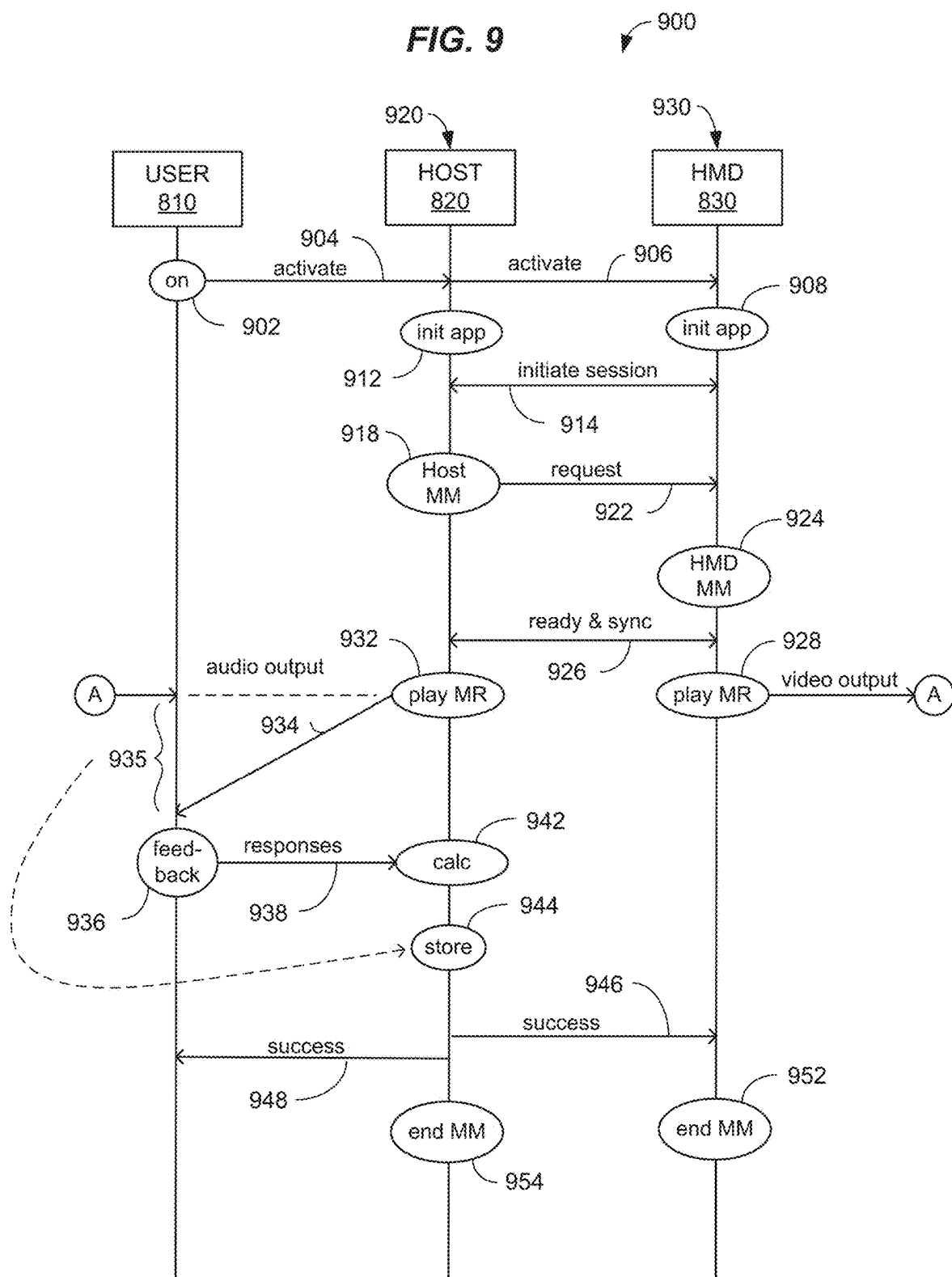
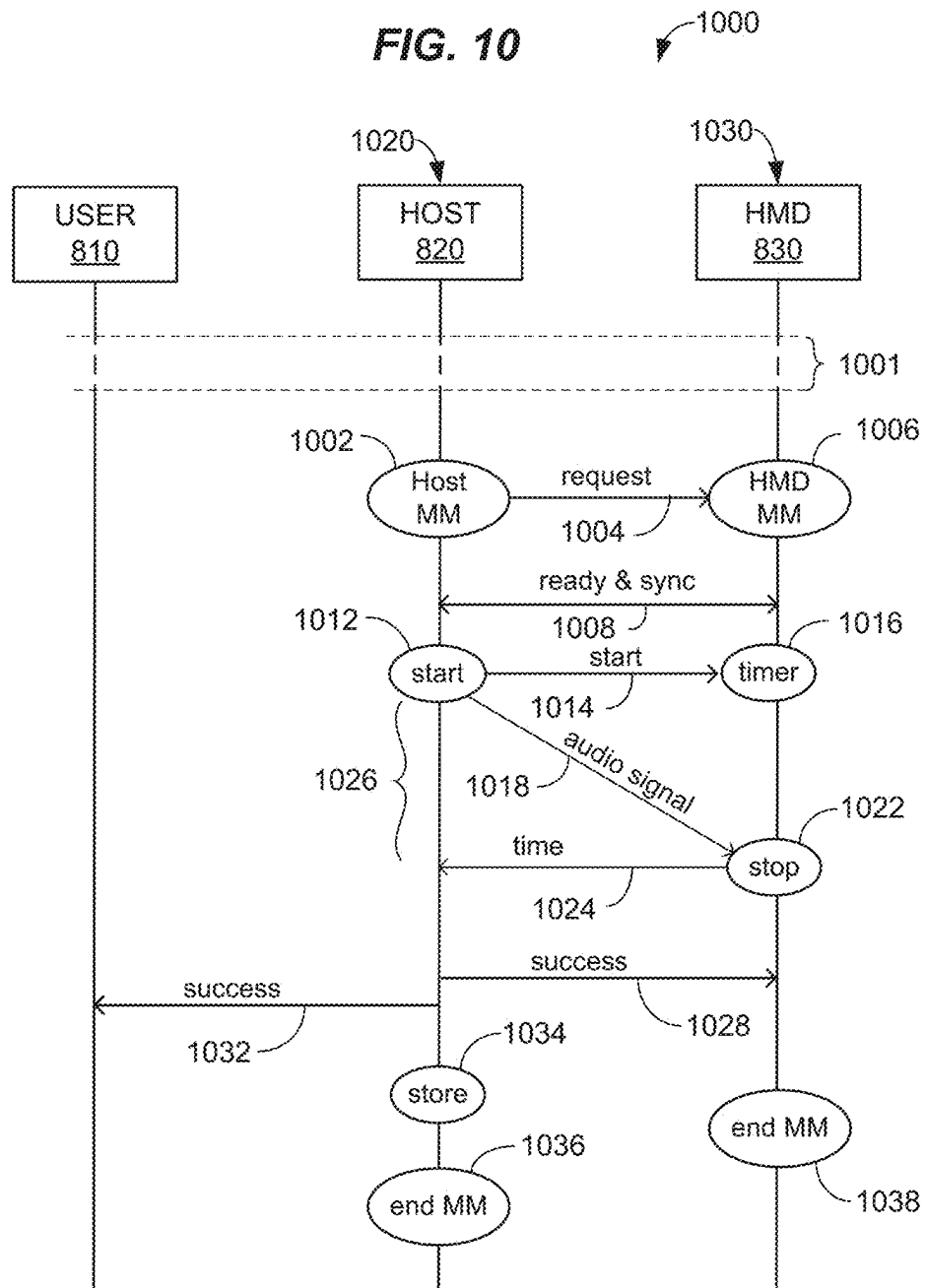
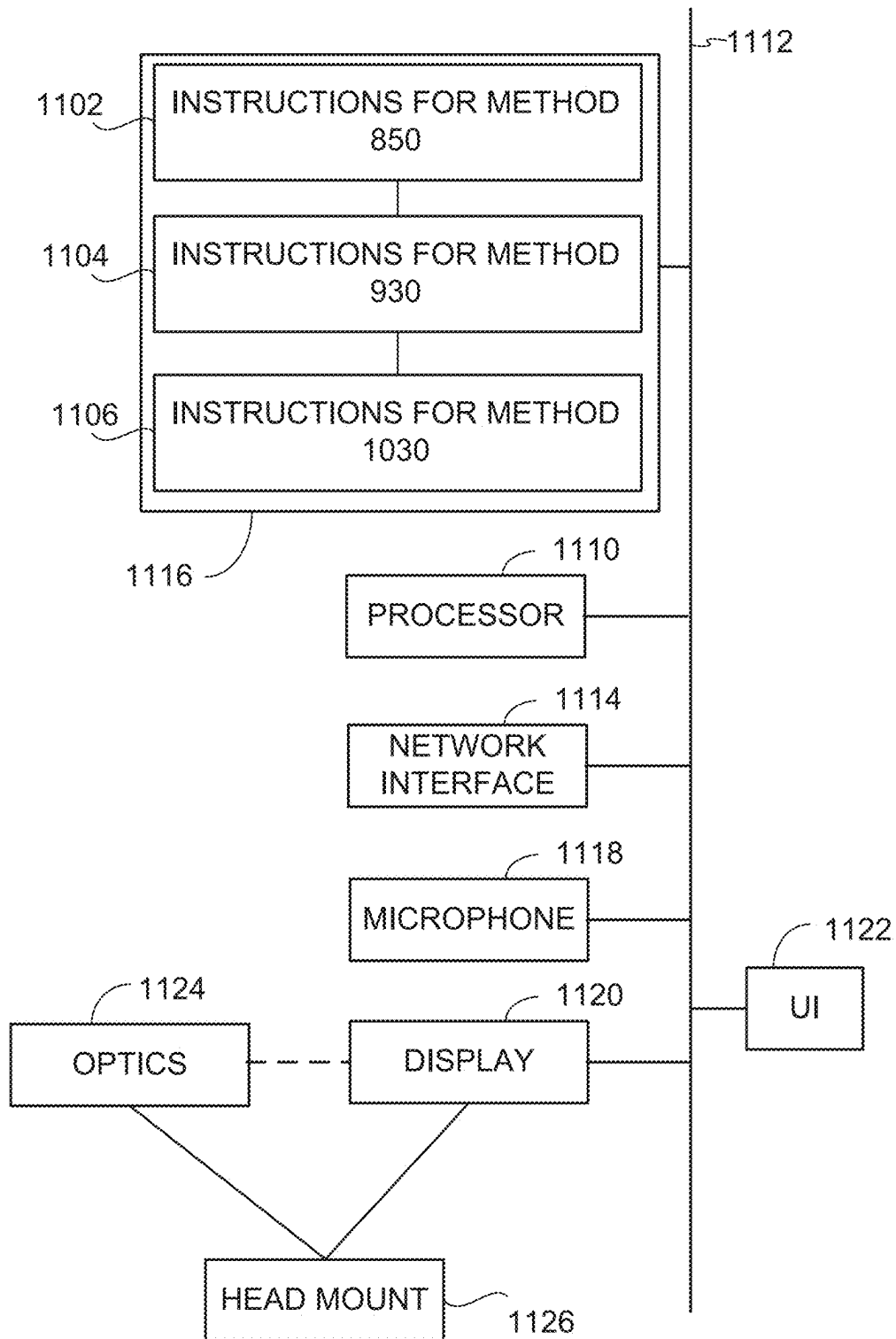
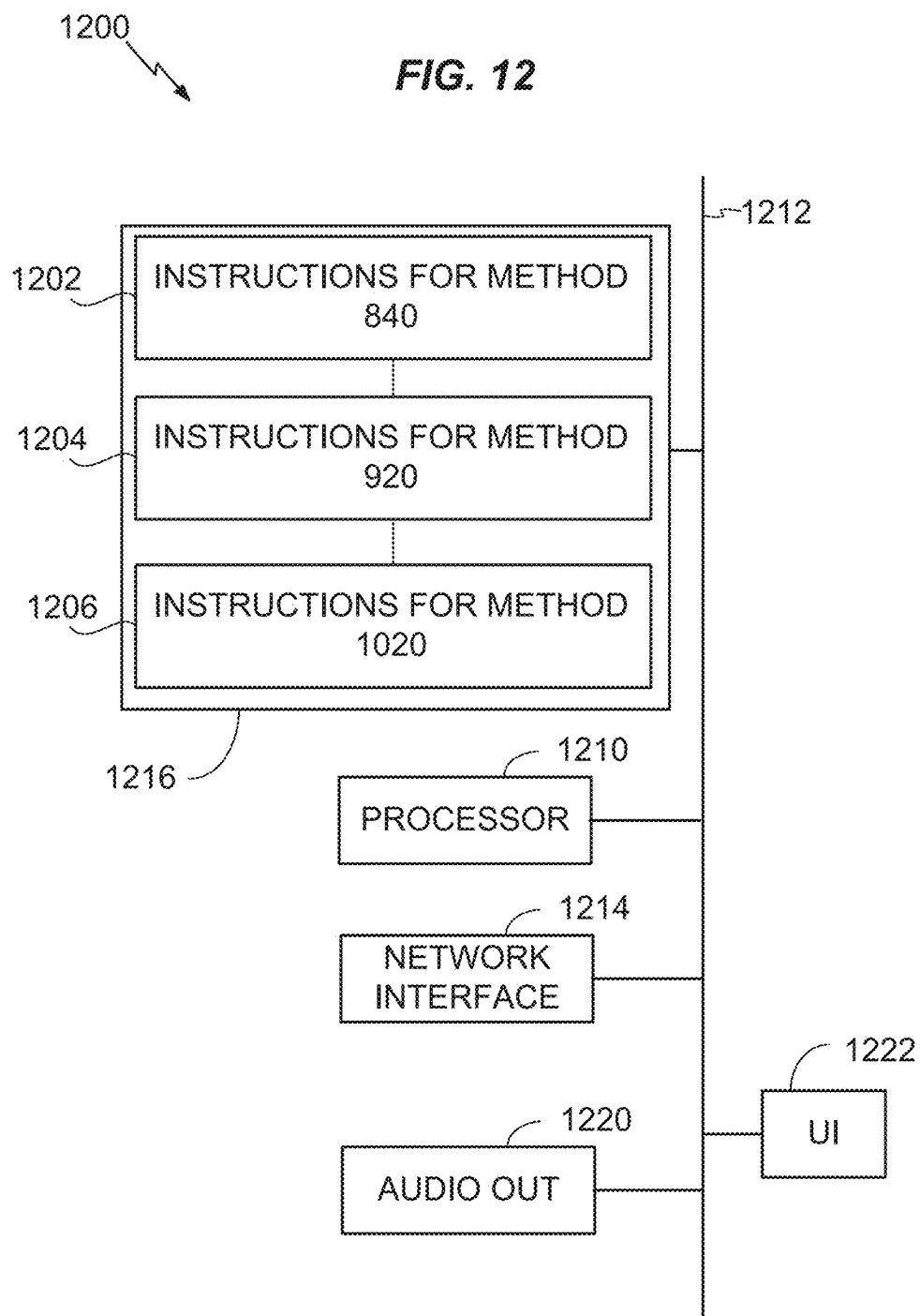


FIG. 10

1100

FIG. 11



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LATENCY COMPENSATION BETWEEN COORDINATED PLAYER DEVICES

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application claims priority to U.S. Provisional Patent Application Ser. No. 63/419,335 filed Oct. 26, 2023, which is incorporated herein by reference.

FIELD

The present application relates to systems, methods, and apparatus for latency compensation between coordinated player devices for example in providing cinematic audio and video for virtual reality (VR) playback.

BACKGROUND

The audio component of VR is too often relegated to substandard headphone audio or inferior audio playback in the head-mounted VR display itself. For example, current virtual reality systems are not integrated into the typical consumer's home theater entertainment system, preventing VR consumers from enjoying a virtual reality experience with the cinematic immersive sound they enjoy in their home systems.

In addition, video quality in VR playback is compromised and inconsistent. Playback quality for VR in a head mounted display (HMD) player device, sometimes referred to herein as an 'HMD,' is inconsistent among models of HMDs, and even among VR video player applications within the HMD. These inconsistencies present a problem for VR content creators as to what compression, video specs, and color profiles to deliver. Additionally, inconsistency in player quality and specifications often requires multiple deliverables for each VR system or playback application to achieve consumer satisfaction.

HMD manufacturers can supply a higher quality of playback by 'tethering' or cabling the HMD to a higher end and higher cost PC, thereby isolating the user from their typical (non-PC) home entertainment products. This lack of wireless capability adds inconvenience, cost and complexity resulting in an inferior user experience and diminished ease of use. In addition, when tethering is used, multiple separately tethered users will have difficulties with cables when sharing the same VR entertainment experience together as they would watching a movie or streaming program.

It would be desirable, therefore, to develop new methods and other new technologies for providing cinematic audio in conjunction with virtual reality (VR) playback, that overcomes these and other limitations of the prior art.

SUMMARY

This summary and the following detailed description should be interpreted as complementary parts of an integrated disclosure, which parts may include redundant subject matter and/or supplemental subject matter. An omission in either section does not indicate priority or relative importance of any element described in the integrated application. Differences between the sections may include supplemental disclosures of alternative embodiments, additional details, or alternative descriptions of identical embodiments using different terminology, as should be apparent from the respective disclosures.

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In an aspect of the disclosure, a virtual reality wireless immersive audio and video player may be embodied as a multi-application software system or method which addresses the deficiencies noted above. Though it is intended as a consumer facing system for the home, it can be used for professional location based virtual reality venues and presentations.

In an aspect, a method for synchronizing audio output by an external sound system driven by a host player device with video output from a head-mounted display (HMD) player device may include establishing a session between the host player device and the HMD player device, wherein the host player device is coupled to an external sound system and the HMD player device comprises a video output system. The method may further include determining, by at least one of the host player device or HMD player device, a latency period between audio output from the external sound system and corresponding video output from the HMD player device, wherein the determining is based on both the host and HMD player devices simultaneously initiating play of test content. The method may further include providing, by the host player device, both the host and HMD player devices with access to identical digital audio-video content selected by a user, and initiating play of the digital audio-video content by the host player device for audio output by the external sound system. The method may include initiating play of the digital audio-video content by the HMD player device after waiting for the latency period so that audio output by the external sound system is synchronized with video output from the HMD player device.

In an aspect, the method may include determining the latency period based on user input received by the host player device in response to contemporaneous output of the test content from the host and HMD player devices. In an alternative, or in addition, the method may include determining the latency period based on detecting, by at least one of the host player device or HMD player device, a lag between audio output from the external sound system and corresponding video output from the HMD player device, wherein both the host and HMD player devices simultaneously initiate play of the test content. In an aspect, the host player device sends a wireless signal to the HMD player device to instruct initiation of the corresponding video output. The signal may include at least one of an audio signal or an optical signal emitted by the external sound system.

In another aspect of the method, establishing a session may further include verifying a connection between first and second applications. The establishing may be done over a WiFi or other suitable network. The method may further include signaling by the HMD player device to the host player device that the HMD player device is ready to play content.

The method may further include ensuring, by the host player device in cooperation with the HMD player device, that the host player device and the HMD player device contain identical copies of the digital audio-video content. This may further include querying by the host player device whether the memory of the HMD player device contains a copy of the digital audio-video content, and copying the content to be played from the host player device to the HMD player device.

The method may include causing deletion after play, by at least one of host player device or the HMD player device, of the digital audio-video content on the HMD player device based on satisfaction of a predetermined condition. The predetermined condition may include at least one of: number of plays, a time-period, or a license type.

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The method may include displaying by the host player device at least one of: a navigation screen with playback controls and latency compensation, video output of the digital audio-video content, overlaid navigation controls, or a 3D video output version of the digital audio-video content. In an aspect, the external sound system may include a home theater system.

The system or method may incorporate specific content workflow and preparation of the VR media files; providing consistency of image quality by specifying the file video and audio deliverables ideally matched to playback software and HMD; supporting wireless HMD player device video playback that can be used simultaneously by multiple users with low bandwidth requirements; and synchronized full immersive home theater and cinema style audio played on a consumer's home entertainment audio system, sound-bar or other audio device. The system or method can be used with speakers, open air or bone induction headphones or a hybrid combination of speakers in the room and integrated audio speakers in the HMD player device for enhanced audio effects.

A system for practicing the method as described herein may sometimes be referred to as an "AudioSync"™ system. As used herein, a 'player device,' 'host device' or 'server' may include at least a computer processor coupled to a memory and to one or more ports, including at least one input port and at least one output port (e.g., a desktop computer, laptop computer, tablet computer, smartphone, PDA, etc.). A computer processor may include, for example, a microprocessor, microcontroller, system on a chip, or other processing circuit. As used herein, a 'processor' means a computer processor. A client device, host device or server may be configured as a 'player device' that outputs audio and/or video output by playing digital audio-video content.

To the accomplishment of the foregoing and related ends, one or more examples comprise the features hereinafter fully described and particularly pointed out in the claims. The following description and the annexed drawings set forth in detail certain illustrative aspects and are indicative of but a few of the various ways in which the principles of the examples may be employed. Other advantages and novel features will become apparent from the following detailed description when considered in conjunction with the drawings and the disclosed examples, which encompass all such aspects and their equivalents.

BRIEF DESCRIPTION OF THE DRAWINGS

The features, nature, and advantages of the present disclosure will become more apparent from the detailed description set forth below when taken in conjunction with the drawings in which like reference characters identify like elements correspondingly throughout the specification and drawings.

FIG. 1 is a schematic diagram illustrating an overview of a virtual reality wireless immersive audio and video player system.

FIG. 2 is a schematic diagram illustrating a system modified for multiple HMD player devices.

FIG. 3 is a schematic diagram illustrating virtual reality media file detail and workflow for a system as described herein.

FIG. 4 is a schematic diagram illustrating aspects of the system for an immersive audio speaker and binaural hybrid.

FIG. 5 is a flow diagram illustrating a method for latency correction in a system 100 with streaming device or game console as source.

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FIG. 6 is a flow diagram illustrating alternative aspects of the method for media loading and synchronization in a virtual reality wireless immersive audio and video player system as described herein.

FIG. 7 is a flow chart illustrating alternative aspects of the method for media management and playback in a virtual reality wireless immersive audio and video player system as described herein.

FIG. 8 is a sequence diagram illustrating alternative aspects of the method for media management and playback in a system as described herein, and methods performed by components of the system.

FIG. 9 is a sequence diagram illustrating a method for determining a latency value for use in playback management in a system as described herein, and methods performed by components of the system.

FIG. 10 is a sequence diagram illustrating an alternative method for determining a latency value for use in playback management in a system as described herein, and methods performed by components of the system.

FIG. 11 is a conceptual block diagram illustrating components of an apparatus or system for a host player device.

FIG. 12 is a conceptual block diagram illustrating components of an apparatus or system for an HMD player device.

DETAILED DESCRIPTION

Various aspects are now described with reference to the drawings. In the following description, for purposes of explanation, numerous specific details are set forth to provide a thorough understanding of one or more aspects. It may be evident, however, that the various aspects may be practiced without these specific details. In other instances, well-known structures and devices are represented in block diagram form to facilitate focus on novel aspects of the present disclosure.

Referring to FIG. 1, an AudioSync Director Virtual Reality video and audio player system 100 enables a user or multiple users to watch and experience 2D 360 or 180 3D or other virtual reality video on a head-mounted display (HMD) player device 102 synchronized with immersive multi-channel audio 104 which includes but need not be limited to Dolby Atmos, DTSX, MPEGH, Auro3D, or other current or future immersive audio formats. The VR media file is played back on an audio system 106 capable of playing surround or immersive audio, such as a multi-channel Dolby Atmos/DTSX/MPEGH home theater system or sound-bar. This system 106 may be configured for use in a consumer home theater entertainment system, professional cinema or location based VR venue, for example. The host playback device 108 may include but need not be limited to Apple computers (MacOS), Android, Linux or Windows based computers, tablets, phones, gaming consoles or streaming devices.

While playback on a wireless HMD player device 102 is preferred, the system can also be implemented for a single user with tethered AR/VR head mounted displays or glasses systems, including but not limited to Magic Leap, NReal, Sony PS5, Varjo and other current and forthcoming devices.

The system 100 may include various components, including, for example: (1) The VR video/audio content 110, 110'—production and customization; and (2) The AudioSync Director playback software application 112 on a 'host device'. A host device may be, or may include, a computer, game console, streaming device, tablet or phone. The components may include (3) an AudioSync playback companion

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application **114** installed on a player device, for example, on an HMD **102** for playing VR content, and (4) a delay compensation method and system implemented by a combination of the AudioSync Director application **112** and AudioSynce playback application **114** in cooperation with the host player device **108** and HMD player device **114**. ‘Delay’ may also be referred to herein as ‘latency.’

The AudioSync Director application **112** currently exists as software for Mac OS, Android, OpenXR and Meta Quest 2, Meta Quest 3 and Meta Quest Pro HMD player devices and may also be deployed to Windows or Android based systems, Linux, other proprietary OS systems and game consoles, tablets, phones or streaming devices, sometimes herein referred to as ‘host devices.’ The HMD companion application **114** can be used on any HMD player device with tethered or wireless capability including but not limited to Meta Quest devices, Pico, Apple or other consumer or enterprise HMD VR devices **102**. It can be used on tethered single user devices that include but not limited to Magic Leap, Varjo, Nreal and other AR/VR devices.

As shown in FIG. 2, a multiple-HMD system **200** may be configured similarly to the previously described single-HMD system **100**, but with multiple HMD player devices **202** all connected to the same host player device **108**. Each of the HMD player devices **202** may be synchronized to the host player device **108** and to each other using the configurations and methods disclosed herein for a single HMD player device **102**.

VR Video Content: Referring to FIG. 3, the system may be designed to use an industry standard format file **110**, for example, .mp4 or .mov or other commonly supported video file that uses an industry standard h.264 or h.265 video or other industry accepted video codec. The video image can be a “top/bottom” or “side by side” 3D video image with a 360 or 180 degree field of view. Such video files for immersive VR content may sometimes be referred to herein as “digital audio-video content.”

The virtual reality video file **110** may be created as a traditional industry standard .mp4 or .mov file with specific video specifications as to bit-rate, color profile, file size and bandwidth to be consistent with playback specifications for the companion video playback application **114** on the HMD player device **102**. This ensures highest quality video image playback consistency and quality desired and designated by the content creator, providing an enhanced user experience.

An immersive or surround multi-channel soundtrack is created for the VR video presentation (movie). This includes but is not limited to surround channel formats such as quad 4.0, 5.0, 5.1, 6.0, 6.1, 7.1 and immersive channel or object based audio formats such as 7.1.2, 7.1.4, 9.1.2, 9.1.4, currently up to 22.2. There is no limit to the number of channels or speakers used.

A custom immersive soundtrack **116** (Dolby Atmos as a typical example) may be created for the VR video presentation. This audio soundtrack file **116** may be then conformed to the VR video file to create a customized virtual reality cinematic experience with full immersive multi-channel audio. This audio portion of the VR video may be deployed as an immersive soundtrack (typically 7.1.4) including but need not be limited to Dolby Atmos, DTSX, MPEGH, Auro 3D or CX or other currently existing or future immersive audio bitstream formats. The audio may be delivered by an audio codec including but not limited to Dolby Digital Plus+jocs (joint object coding) “Atmos” bitstream. The immersive soundtrack may be output by an external sound system **106** (FIGS. 1-2).

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As an example: The Dolby Atmos audio **302** is industry standard, up to 7.1.4 channels and must be created and produced with standard Dolby Atmos audio production workflow using (but not limited to) Dolby’s or other currently available software tools from Avid, Apple, Yamaha or other audio software developers now and in the future. The final Atmos mix is then exported via Dolby’s software toolset or toolset integrated into the (DAW digital audio workstation software) to the Dolby Digital Plus+joc bitstream as a final deliverable. This bitstream is then conformed to the VR video file to create a VR video file with Dolby Atmos, DTSX, Auro 3D or CX or other immersive audio.

Referring to FIG. 4, a system **100** may include the VR video file **110** configured to have two soundtracks. The primary soundtrack **116** is the immersive audio track as described above. This typically is the first audio stream, herein referred to as primary stream #1, for output by the external sound system **106** or soundbar. The second soundtrack **118** may be a stereo binaural stream using a commonly used audio codec such as MP3 or AAC or other widely used stereo audio stream. This stereo stream can be either a binaural version of the immersive soundtrack in primary stream #1 that users can listen to on headphones with the HMD player device or on small speakers on the HMD player device **102** itself. This stream shall herein be referred to as ‘secondary stream #2.’

Another use of the secondary stream #2 as stereo or binaural audio stream **118** may include a ‘hybrid’ method where additional audio effects, sounds or audio enhancements can be played simultaneously through the built-in speakers of the HMD, or headphones along with the immersive audio system playing in user’s room via their home theater or sound-bar. This secondary stream of audio may sometimes be referred to herein as “auxiliary audio” content.

Referring again to FIGS. 1-2, the VR video file **110**, **110'** may be duplicated on the HMD player device **102** by copying it from the host player device **108** and application **112** and stored in local storage on both the host player device and the HMD player device. The files may also be stored and utilized on external storage, for example, external hard drives, SD cards or USB drives. The AudioSync Director software application **112** may be located on the host player device **108**, for example, a computer, game console, tablet, phone or streaming device.

The AudioSync companion software application is located on the HMD player device **102** or on each of the multiple HMD player devices **202**. Both the host player device **108** and each HMD player device **102** may be connected to and logged on to the same WiFi network **120**, or equivalent local area wireless network.

The applications **112**, **114** may be configured such that when both the AudioSync Director application **112** and the companion application **114** on the HMD player device **102** on the same wireless network **120** and launched, the two applications detect each other on the network. The companion application may output an indication for output on a display of the HMD player device **102** that the companion application **114** is connected to the application **112** on the host player device **108**, establishing a play session in which the applications **112**, **114** will cooperate to synchronize audio output from the external sound system **106** with video output from the HMD player device **102**.

Referring to FIGS. 5-7, aspects of a method **400** for synchronizing audio output by an external sound system **106** driven by a host player device **108** with video output from an HMD player device **102** are shown.

A VR video file **110** containing digital audio-video content may be selected for the first time by user interaction with the AudioSync Director software **112** on the host player device **108**. The following occurs:

The AudioSync Director application **112** creates a duplicate **110'** of the file **110** in its directory/local library folder. The copy **110'** is created with a unique renaming of the file.

When it is launched, the companion application **114** on the HMD player device **102** connects to the same Wi-Fi network **120** as the host player device **108**. The host player device determines **410** (FIG. 7) whether a copy of the digital audio-video content **110** exists on the HMD player device **102**. If the copy **110'** exists, the host player device **108** does not make a copy **412**. If the files does not exist, the host player device renames and copies **414** the content file as previously described herein.

Once connected to the host player device **108** via the network **120**, the HMD player device **102** detects the file **110'** containing the digital audio-video content and immediately copies the file with unique re-naming to local storage **103** in the HMD player device **102** over the Wi-Fi network **120**. This duplicates the VR file **110'** on the HMD as a uniquely renamed copy of the original file **110** contained in local storage **109** of the host device.

The companion application **114** reports the progress of copying or 'downloading' the file on the screen of the HMD player device **102**, visible to the user. Once copying the file to the HMD player device is complete, the companion application on the HMD player device visually reports the completion of copying of the file and indicates on the HMD player device screen that the digital audio-video content is ready to play.

The file copying process between the host device application **112** and the companion application **114** on the HMD player device occurs the first time the file is selected by the user in the application on the host device because the file copy **110'** is not present on the HMD. If the file is already present in the directory of the HMD player device **102**, the copying process does not happen again unless and until the user or a system component deletes the file **110'** from local storage of the HMD player device.

Once the file is stored locally on the HMD player device, and the user initiates start/play on the host device **108** by interacting with the AudioSync Director application **112**, playback of the file **110** commences immediately on the host player device **108** and simultaneously preparation for playback of the duplicate file **110'** commences on the HMD player device **102**.

The companion application **114** detects the uniquely named file **110'** from the host player device **108** that has been requested to play via the AudioSync Director application **112**. As soon as the companion application **114** detects a 'start' sync signal **402** (FIG. 6) from the host player device **108** over the WiFi network, the HMD player device **102** initiates playback of the identical duplicate file **110'** from local storage. Thus, the 'start' signal from the AudioSync Director application **112** triggers initiation of playback of the VR video file **110'** on the HMD player device **102**. The 'start' sync signal is generated after applying **404** (FIG. 6) a latency compensating wait period by the AudioSync Director player application **112** on the host player device **108**. The AudioSync Director application **112** sends the 'start' sync signal sent via the wireless network **120** to the companion application **114** on the HMD player device **102**. Thus, playback of the VR video file on the HMD player device **102**

is synchronized with immersive audio output by the external sound system **106** from playback of the content by the host player device **108**.

The latency arises from transmission and output processing between the host player device **108** and the external sound system **106**. The host player device **108** may be connected to a user's immersive audio capable system **106** or sound-bar via HDMI or other digital output. The AudioSync Director application **112** plays the VR video and outputs **406** (FIG. 6) an immersive audio bitstream **408** from the HDMI or digital output of the host player device **108**. The routing of the digital immersive audio stream and subsequent decoding and processing in the external audio system **106** or sound-bar causes latency. The image and sound will not be synchronized. The audio will appear late to the picture, if not corrected. The latency may usually be less than one second (e.g., 10-100 ms) depending on the system, enough to be perceivable by the user. As a result, while watching the video portion of the VR file in the HMD, the audio output by the external sound system will be delayed, late and out of sync with the image in the HMD player device.

Applying latency compensation **404** during playback eliminates perceivable synchronization error in the systems **100**, **200** and is described below:

To remedy latency and bring the audio portion playing from the host player device **108** into sync with the video portion playing from the HMD player device **102**, latency compensation is programmed into the AudioSync Director application **112** operating on the host device **108**.

The latency compensation may include a delay built into the AudioSync Director application **112** on the host player device that delays the play/start signal sent to the companion application **114** on the HMD player device **102** over the wireless network **120**. This delays the playback of the file **110'** on the HMD player device by several milliseconds later than the playback of the file **110** on the host device.

Consequently, the video portion of the VR file **110'** starts slightly later on the HMD player device **102** than the playback of the corresponding file **110** by the host player device **108**, which compensates for the delay caused by additional routing to HDMI and decoding/processing of the audio in the host device **108** and external audio system **106**. Thus, the video output from the HMD player device and the immersive audio output from the external system **106** or sound-bar are synchronized.

The amount of latency compensation in the AudioSync Director application **108** on the host device may be variable and adjustable by the user or automatically by the system. This allows adjustability in the event the user has any given variations of latency that may occur in HDMI cabling, audio decoding or processing in the external audio system or device.

The user initiates **401** (FIG. 7) play of VR video file with immersive audio track on the AudioSync Director application on the host device. The sync signal **402** with latency compensation **404** applied is provided from the host player device **108** over a wireless or wired connection to the HMD player device **102**.

The companion application **114** detects the start sync signal and begins playback of the identical video file on the HMD player device. Both files begin to play synchronized with the immersive audio stream decoded and synchronized on the user's external immersive sound system **106**.

Multiple HMD player devices **202** can be used wirelessly with the same time playing the same program material

simultaneously as described above if all have the companion application 114 installed on each of the multiple HMD player devices.

AudioSyncDirector Software Application: The AudioSync Director Software application 112 for Mac OS, Windows, Linux, Android, OpenXR or other software video player may have the following functions:

Selection and playback of the VR video file with the immersive audio soundtrack outputting the immersive audio bitstream including but not be limited to Dolby Atmos, DTSX, MPEGH, Auro 3D or CX or other current or future formats from the digital output or HDMI output of the device including but need not be limited to Apple computers (Mac OS), PC, Android, OpenXR, Linux, game console, tablet, phone or streaming devices. These are known herein as 'host devices'.

Once the VR video file is selected and loaded, the AudioSync Director application manages the transfer of the VR video file the user wishes to play with the AudioSync companion application to local storage of an HMD player device. The transfer may be executed over a wireless local area network that both the host device and HMD player device are logged on to. File copying may also be executed across a wired system (also known as a tethered system). When the application on HMD detects the application on the host device and the VR video file selected and is loaded by the user, the file may be automatically copied to the local storage of the HMD where it is played by the companion application on the HMD.

Once the file is copied into the local storage of the HMD it does not have to be copied again in the future unless it is deleted from local storage on the HMD. If the video file is present on the HMD and the user selects the file again on the AudioSync Director application—the companion application on the HMD will launch it and be ready to play it. A message confirming may be displayed on the HMD display screen.

Upon starting playback of the VR video file on the AudioSync Director software, the application generates a start/sync signal delayed a few milliseconds (providing latency compensation) from the application on the host device to the HMD companion app commencing the synchronized playback of the identical video file on the HMD. The video portion on the HMD plays in sync with the audio output from the host device and external audio system.

The immersive audio bitstream of the VR video file is output from the host device via the digital audio outputs of the host device including but need not be limited to HDMI, Thunderbolt or USB-C. This immersive audio bitstream may be routed to the HDMI input or digital input of the user's immersive audio capable sound system or sound-bar for decoding, processing and audio playback.

This method and system can be used in a professional cinema or location based VR environment or venue or home entertainment environment.

AudioSyncStation Companion App for HMD player devices: The AudioSync companion app 114 for is an application for any VR/AR HMD player device that is configured to cooperate with the AudioSync Director application 112 on a host device in that it will play the .mp4 video described in VR Video content above, or other suitable file format. The functionality and use may be as follows:

The companion application is installed on the user's HMD. The HMD and host device are connected to and logged into the same wifi network. The user launches the applications on both the host device and the HMD.

The AudioSync Director application on the host device and the companion app on the HMD player device detect each other over the same communication network.

The user selects a VR video to play on the AudioSync Director app on the host device. A duplicate video with different naming configuration is copied into the directory/library folder of the AudioSync Director app on the host device.

The companion application detects the user has selected a video on the AudioSync Director application. If the video file does not already exist in the directory of the companion application on the HMD player device, the Host device application immediately uploads the video file the user has selected on the application on the host device and copies the file via a wireless or wired connection to the local storage (or external storage) on the HMD.

This copying progress may be displayed on the screen of the HMD while in process. When copying/uploading of the file to the HMD player device is complete, the companion app may display an onscreen message in the HMD device that the file is present and ready for playback.

The companion application on the HMD awaits a start play signal from the AudioSync Director application on the host device. Once play is initiated on the host device application, a latency compensated start/play signal is sent over the wireless network to the companion app in the HMD player device. The start/play signal sent to the companion application is delayed by a few milliseconds to compensate for latency caused in the host device and external audio system. Once playback is initiated on the AudioSync Director application both videos on the host device and the companion app begin to play in synchronization—the image in the HMD and the audio from the host device and external audio system.

The companion application on the HMD can play the primary immersive audio stream (stream #1) of the VR video file if that type of audio file is supported on the HMD player device and playback software. Users can play this option through the speakers on the HMD or headphones if they choose this option.

The companion application on the HMD will play an industry standard stereo or binaural stream as audio stream #2 through the built-in speakers of the HMD or with headphones if the user chooses this option.

Example: In-Use Case: The user connects their host device via HDMI to the immersive audio system of their home entertainment system capable of playing back Dolby Atmos, DTSX, MPEGH, Auro 3D or CX or other audio streams on an immersive compatible home theater system or sound bar.

The AudioSync Director software application is installed on the user's host device.

The user has an HMD player device with the AudioSync companion application installed on it.

Both the host player device and the HMD player device are connected and logged on to the same local network.

A VR video file with an immersive audio stream is selected by the user and loaded on to the AudioSync Director application on the host device either locally or downloaded from a server or cloud.

The user installs and launches AudioSync companion app on their HMD player device.

The host application and companion application detect each other on the same local network.

When the video file is selected on the host device application, the companion app detects whether it has a copy of that file in its directory in local storage on the HMD. If it

does not, the host device app immediately copies the video file with a re-naming configuration from the directory of the host device application across WiFi to the HMD companion application and the local storage (or external storage) of the HMD player device.

The HMD player device displays the copying/downloading status on its screen. When copying is complete, the companion app displays a message that the file copying is complete, and the file is ready to play.

There are now identical digital audio video files on both the computer or host device application and on the HMD.

If the identical file already exists in the directory in local storage of the HMD, the companion app selects it and displays a message on its display that the file is ready to play.

The companion application awaits a start/play command from the host device.

For first use, the user either A) runs a test digital audio video file that plays audio and video test signals allowing the user can see and hear the latency between the audio outputting from the external audio system and the video displaying in the HMD device. The user adjusts the latency compensation control on the host device application (AudioSync Director) using the host device remote control to adjust the latency compensation value displayed on the video monitor or TV connected to the Host device until the audio signal outputting from the external audio system is in sync with the video signal displayed in the HMD device. Or B) the user initiates an automatic latency compensation software routine where a test digital audio video file or software in the host device generates an audio test signal from the host device and external audio system that is detected by the microphones built into the HMD device and determines the lag or latency of the time the audio signal arrives at the HMD, calculates the latency compensation value, sends that value back to the host device and sets the latency compensation setting on the host device player application. Or alternatively, C) the user initiates an automatic latency compensation software routine where a test digital audio video file or software in the host device generates a video test signal from the host device that is displayed on a video monitor or TV connected to the host device that is detected by the cameras built into the HMD device and determines the lag or latency of time the video signal arrives at the HMD cameras from the host device screen, calculates the latency compensation value and sends that value back to the host device and sets the latency compensation setting on the host device player application. Or the user initiates an automatic latency compensation setting routine that is a combination of B and C. Once the latency compensation value is set on the host device player application, the files are ready to play.

The user initiates start/play on the host device application and the AudioSync Director application generates a start/play signal with latency compensation applied to HMD that is delayed by a few milliseconds to compensate for audio delay caused by the host device and external audio system.

Both video file will begin to play simultaneously on both the host device and the HMD. The VR video image in the HMD will play synchronized with the immersive audio portion playing out of the HDMI output of the host device into the HDMI input of the Dolby Atmos, DTSX, MPEGH, Auro 3D or CX or other immersive or surround compatible audio device or system.

The AudioSync Director application on the host device compensates for any latency caused by audio signal flow, decoding and processing in the external audio device so that the audio is in sync with the video playback in the HMD.

The user now enjoys a virtual reality narrative experience with full surround or 7.1.4 or higher immersive cinematic audio with the video image component playing in sync in the HMD wirelessly. Multiple users can enjoy the same experience with multiple HMD's with the companion app installed on all of them. A single host device can drive and connect to multiple HMDs.

If the user decides they do not wish to use their home entertainment immersive speaker system—the immersive soundtrack can be played from their HMD player device from the primary immersive stream #1, if that audio format is supported, or secondary stereo or binaural secondary audio stream #2 in the VR video file on headphones for from the speakers on their HMD. This use case could be for late night or private listening.

As another option—the user is enjoying their VR experience with their immersive home theater sound system—but secondary track #2 in the VR video file contains additional enhanced sound effects that can be played through the speakers built into the HMD at the same time as the full immersive audio presentation on the users home theater system giving the user an enhanced VR audio experience with proximity effects utilizing both the speaker system and built-in speakers in the HMD.

Alternative approaches for playback of a narrative cinematic VR experience as described may include using VR video playback software on a host device to stream the video and audio via a network to a companion application on the HMD. Disadvantages of this approach may include high bandwidth requirements, lack of support for synchronization or utilization of actual immersive audio or external surround audio, limitation of audio output to the HMD via a cable tethered or Bluetooth which have latency issues resulting in complexity and poor user experience, and restriction to a single user. Multiple HMDs could not easily be used with stable synchronization of image or sound therefore only a single user can enjoy the experience.

Another approach might include uploading the VR Video to the cloud for later download to synchronized HMD devices driven by a computer or tablet content management system such as the Headjack system (<https://headjack.io/>). Disadvantages of this methodology may include: high bandwidth needed for uploading and downloading to a cloud based system and corresponding delay; limitation to content management systems processing in the cloud which compromises the video content and force users into lower quality versions and a lower quality experience; synchronization between HMD's and control devices is not stable and can exhibit latency issues, and immersive audio is not supported.

The present system and methods do not suffer from these disadvantages.

In accordance with the foregoing, and by way of further explanation, FIG. 8 shows certain illustrative aspects of the foregoing methods in a system as described herein, including methods **840**, **850** performed by components of a system including a host player device **820** and HMD player device **830**, and a combination method **800** interacting with a user **810** of both player devices **820**, **830**. At **802**, the user **810** interacts **802** with an 'on' feature of a user interface, activating **804** the host player device **820**. The 'on' feature may be incorporated in the host device **820** or in the HMD player device **830**. At **806**, the host player device **820** initiates a playback management application **806**, for example an AudioSync Director application as described herein above, which may display or otherwise output **808** a menu interface for user selection of digital audio-video content for contemporaneous play by the host player device

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820 and the HMD player device **830**. The method **840** performed by the host device **820** may be controlled or initiated by the playback management application.

At some time prior to beginning the contemporaneous playback, shown here at **812**, the user may activate **814** the HMD player device, for example by putting it on and activating a switch. One the HMD player device **830** is activated by the user, it may load into its memory and initiate execution **815** of a companion playback management application, for example a companion application configured to work with an AudioSync Director application, which may enable the HMD device **830** to perform operations of the method **850**.

The user **810** may access a content selection interface served by either or both of the host player device **820** or the HMD player device **830**. At **816**, the user selects an audio-video title to be played, which title may be contained in a digital audio-video content as described herein above, causing an indication **818** of a title selected for playback to be provided to the host player device **820**. At **822**, if the digital audio-video content for the selected title is not already present in its local memory, the host player device **820** may download the content from a content server (not shown). At **824**, the host player device **820** in cooperation with the HMD player device **830** initiate a playback session **824**. Initiating a session **824** may include, for example, establishing a wireless data connection (e.g., Wi-Fi or Bluetooth), and executing a handshake protocol between the playback management applications on the two player devices **820**, **830**.

At **826**, the host player device **820** may transmit a copy of the digital audio-video content to the HMD player device **830** via the wireless data connection supporting the playback session. Optionally, the host playback device may record relevant metadata concerning the title, destination, and time/date for use in content management, and/or may transmit relevant metadata to a content management server.

At **828**, the host player device **820** may determine a latency value for the HMD player device **830**. The latency value may be a characteristic, stable value that is measured only once or only occasionally, and retained **834** in a memory of the host player device **820** and/or HMD player device **830**. Examples of methods **900**, **1000** for measuring latency are described in connection with FIGS. 9 and 10. Once both players **820**, **830** have copies of the content to be played in their respective local memories and the latency is determined and in a memory of the host player device **820**, the host **820** and HMD **830** exchange signals **832** indicating they are both ready for playback to begin.

At **836**, the user **810** via an interface of the host player device **820** provides a "start play" instruction **838** to the HMD player device **830**. Upon receiving the instruction **838**, the host player device **820** initiates play **841** of the digital audio-video content. Upon receiving the start signal **838**, the host player device **820** starts a timer **842** for a wait period **834** based on (e.g., equal to) the determined latency value. During the wait period **834**, the delay **834** is introduced by signal transmission and processing as described herein above.

At **844**, the wait timer **842** expires and the host player device **820** sends a signal via the wireless data connection to the HMD player device **830** instructing the HMD player device to begin playback. Upon receiving the start signal, the HMD player device initiates playback, displaying video output **848** which is visible to the user **810** at the same time as the audio output by the external sound system.

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The synchronization between the audio output by the host player device **820** and the HMD player device may be maintained if the user **810** pauses the playback, as shown beginning at **854**. Here, the user activates a pause feature of the HMD device's user interface, causing it to receive **856** a "pause" instruction. Upon receiving the pause instruction, the HMD device **830** pauses playback and contemporaneously signals a pause instruction **862** to the host player device **820**. The HMD player device **830** maintains the pause position in its memory **864**, while the host player device **820** maintains the pause position **868** in its memory. Thus, synchronization of the audio and video is maintained during the pause.

At **872**, the user **810** activates a "resume" feature **872** of the HMD user interface, sending a resume instruction **873** to the HMD. The HMD resumes play at **874** and contemporaneously sends a resume instruction **876** to the host player device, which may immediately resume audio playback **878** at the stored **868** pause position. In an alternative, if the HMD player device **830** is characterized by additional latency when resuming play after a pause, the host player device **820** may wait for a period based on a determined latency before resuming audio output, as previously described. After play is resumed, synchronized playback by both player devices **820**, **830** continues until the end of the digital content is reached or the user terminates play.

After completing play, the players **820**, **830** may cooperate to close the playback session and enforce any applicable content license or security requirements, as described herein.

FIG. 9 shows a method **900** for determining a latency value for use in playback management in a system as described herein, comprising methods **820**, **830** performed by components **820**, **830** of the system, respectively. At **902**, the user activates **904** the host player device **820**, such as by using a remote control user interface (UI) device. In turn, the host player device activates **906** the HMD player device. At **912** the host player device begins execution of an application, for example the AudioSync Director, that includes a Measurement Module (MM) **918** for latency determination. Likewise, at **908** the HMD player device initiates an application, for example the companion application, that includes a Measurement Module (MM) **924** for latency determination. After initiating the MM at **918**, the host player device requests that the HMD player device begin execution of its measurement module, and the HMD player device does so at **924**. The host **820** and HMD **830** then sync up and signal mutual readiness to begin latency determination, at **926**.

At **932**, the host player device begins playback **932** of a measurement recording (MR) used for latency measurement. Sometime prior to this, the host **820** supplies a copy of the MR to the HMD **830** as previously described for digital audio-video content. The MR may include an introductory portion that instructs the user to provide feedback **936** upon hearing specified audio output from the external sound system after seeing a specified visual event in the HMD **830** output. For example, the instructions might say "press enter as soon as you hear a beep from the sound system after the screen flashes." This feedback **936** provides an indication of the delay **935** in output of the test audio signal **934**. The feedback routine may be repeated several times to obtain an aggregate estimate of the delay **935** perceived by the user. Each time the host **820** receives a user input response at **938**. At **942**, the host may calculate an estimate of the latency value, for example taking an average of the user feedback after adjustments for greater accuracy.

At **944**, the host player device **820** may store the calculated latency value in a non-volatile memory location for use

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during content playback, as described in connection with the methods herein above. At **946**, **948** the host device may inform the user **810** and the HMD **830** that the measurement is successfully completed. At **954**, the host player device may terminate the MM **954**. The HMD player device terminates the MM at **952**.

FIG. **10** shows an alternative method **1000** for determining a latency value for use in playback management in a system as described herein, and methods **1020**, **1030** performed by components **320**, **830** of the system, respectively. The first portion **1001** of the method **1000** may be the same as steps **902-926** of method **900**.

At **1002**, the host player device **820** initiates a measurement module (MM) and instructs the HMD player device **830** to initiate a corresponding HMD-side MM, which the HMD does at **1006**. At **1008**, the host **820** and HMD **830** exchange signals indicating mutual readiness to begin latency determination.

At **1012**, the host **820** starts play of test content and simultaneously instructs **1024** the HMD **830** to start a timer **1016**. At **1018**, the audio output by the host **820** is transmitted and processed through the external sound system, introducing a delay **1026** for reasons explained hereinabove until the audio signal reaches the HMD **830**. At **1022**, the HMD detects the audio signal using its microphone and stops the timer. At **1024**, the HMD **830** send the timer value when stopped to the host player device **820**, indicating the amount of delay **1026**. At **1028**, **1032** the host player device signals that the delay is successful to the user **810** and HMD player device **830**.

At **1034**, the host player device stores the latency value **1026** for later use as described herein and terminates the measurement module **1036**. At **1038**, the HMD player device likewise terminates its measurement module.

FIG. **11** is a conceptual block diagram illustrating components of a virtual reality wireless immersive audio and video player, also called an HMD player device as described herein, according to one embodiment. As depicted, the apparatus **1100** may include functional blocks that can represent functions implemented by a processor, software, or combination thereof (e.g., firmware).

The apparatus or system **1100** may comprise an electrical component **1102** for performing operation of the method **850** shown in FIG. **8**. The component **1102** may be, or may include, a means for said performing. Said means may include the processor **1110** coupled to the memory **1116**, and to the network interface **1114**, the processor executing an algorithm based on program instructions stored in the memory. Such algorithm may include a sequence of more detailed operations, for example, as detailed in FIG. **8** at **850**.

The apparatus or system **1100** may comprise an electrical component **1104** for performing operation of the method **930** shown in FIG. **9**. The component **1103** may be, or may include, a means for said performing. Said means may include the processor **1110** coupled to the memory **1116**, and to the network interface **1114**, the processor executing an algorithm based on program instructions stored in the memory. Such algorithm may include a sequence of more detailed operations, for example, as detailed in FIG. **9** at **930**.

The apparatus or system **1100** may comprise an electrical component **1106** for performing operation of the method **1030** shown in FIG. **10**. The component **1106** may be, or may include, a means for said performing. Said means may include the processor **1110** coupled to the memory **1116**, and to the network interface **1114**, the processor executing an algorithm based on program instructions stored in the

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memory. Such algorithm may include a sequence of more detailed operations, for example, as detailed in FIG. **10** at **1030**.

The apparatus **1100** may optionally include a processor module **1110** having at least one processor, in the case of the apparatus **1100** configured as a data processor. The processor **1110**, in such case, may be in operative communication with the modules **1102-1106** via a bus **1112** or other communication coupling, for example, a network. The processor **1110** may effect initiation and scheduling of the processes or functions performed by electrical components **1102-1106**.

In related aspects, the apparatus **1100** may include a network interface module **1114** operable for communicating with a storage device over a computer network. In further related aspects, the apparatus **1100** may optionally include a module for storing information, such as, for example, a memory device/module **1116**. The computer readable medium or the memory module **1116** may be operatively coupled to the other components of the apparatus **1100** via the bus **1112** or the like. The memory module **1116** may be adapted to store computer readable instructions and data for effecting the processes and behavior of the modules **1102-1106**, and subcomponents thereof, or the processor **1110**. The memory module **1116** may retain instructions for executing functions associated with the modules **1102-1106** and other operations of an HMD player device as described herein. While shown as being external to the memory **1116**, it is to be understood that the modules **1102-1106** can exist within the memory **1116**.

The apparatus **1100** may further include a microphone **1118**, a stereoscopic display device **1120**, and a user interface device **1122** each coupled to the processor **1110** and memory **1116** via the bus **1112** or other coupling. The apparatus may further include an optical train **1124** through which the user views the stereoscopic display **1120** via a mechanical coupling included in a head mount structure **1126**, typically including a case, cushioning, and elastic band.

FIG. **12** is a conceptual block diagram illustrating components of an apparatus **1200** for a host player device as described herein, according to one embodiment. As depicted, the apparatus **1200** may include functional blocks that can represent functions implemented by a processor, software, or combination thereof (e.g., firmware).

The apparatus or system **1200** may comprise an electrical component **1202** for performing operation of the method **840** shown in FIG. **8**. The component **1202** may be, or may include, a means for said performing. Said means may include the processor **1210** coupled to the memory **1216**, and to the network interface **1214**, the processor executing an algorithm based on program instructions stored in the memory. Such algorithm may include a sequence of more detailed operations, for example, as detailed in FIG. **8** at **840**.

The apparatus or system **1200** may comprise an electrical component **1204** for performing operation of the method **920** shown in FIG. **9**. The component **1203** may be, or may include, a means for said performing. Said means may include the processor **1210** coupled to the memory **1216**, and to the network interface **1214**, the processor executing an algorithm based on program instructions stored in the memory. Such algorithm may include a sequence of more detailed operations, for example, as detailed in FIG. **9** at **920**.

The apparatus or system **1200** may comprise an electrical component **1206** for performing operation of the method **1020** shown in FIG. **10**. The component **1206** may be, or may include, a means for said performing. Said means may include the processor **1210** coupled to the memory **1216**, and

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to the network interface **1214**, the processor executing an algorithm based on program instructions stored in the memory. Such algorithm may include a sequence of more detailed operations, for example, as detailed in FIG. **10** at **1020**.

The apparatus **1200** may optionally include a processor module **1210** having at least one processor, in the case of the apparatus **1200** configured as a data processor. The processor **1210**, in such case, may be in operative communication with the modules **1202-1206** via a bus **1212** or other communication coupling, for example, a network. The processor **1210** may effect initiation and scheduling of the processes or functions performed by electrical components **1202-1206**.

In related aspects, the apparatus **1200** may include a network interface module **1214** operable for communicating with a storage device over a computer network. In further related aspects, the apparatus **1200** may optionally include a module for storing information, such as, for example, a memory device/module **1216**. The computer readable medium or the memory module **1216** may be operatively coupled to the other components of the apparatus **1200** via the bus **1212** or the like. The memory module **1216** may be adapted to store computer readable instructions and data for effecting the processes and behavior of the modules **1202-1206**, and subcomponents thereof, or the processor **1210**. The memory module **1216** may retain instructions for executing functions associated with the modules **1202-1206**, and other operations of a host player device as described hereinabove. While shown as being external to the memory **1216**, it is to be understood that the modules **1202-1206** can exist within the memory **1216**.

The apparatus **1200** may further include an audio output port **1120** as described hereinabove for providing an audio signal to an external sound system. The apparatus **1200** may include a user interface **1222** as is conventional in the art.

The various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the aspects disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

As used in this application, the terms “component”, “module”, “system”, and the like are intended to refer to a computer-related entity, either hardware, a combination of hardware and software, software, or software in execution. For example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, a program, and/or a computer or system of cooperating computers. By way of illustration, both an application running on a server and the server can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers.

Program instructions may be written in any suitable high-level language, for example, C, C++, C #, JavaScript, or Java™, and compiled to produce machine-language code for execution by the processor. Program instructions may be

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grouped into functional modules, to facilitate coding efficiency and comprehensibility. It should be appreciated that such modules, even if discernable as divisions or grouping in source code, are not necessarily distinguishable as separate code blocks in machine-level coding. Code bundles directed toward a specific function may be considered to comprise a module, regardless of whether machine code on the bundle can be executed independently of other machine code. In other words, the modules may be high-level modules only.

Various aspects will be presented in terms of systems that may include several components, modules, and the like. It is to be understood and appreciated that the various systems may include additional components, modules, etc. and/or may not include all the components, modules, etc. discussed in connection with the figures. A combination of these approaches may also be used. The various aspects disclosed herein can be performed on electrical devices including devices that utilize touch screen display technologies and/or mouse-and-keyboard type interfaces. Examples of such devices include computers (desktop and mobile), smart phones, personal digital assistants (PDAs), and other electronic devices both wired and wireless.

In addition, the various illustrative logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed with a general purpose processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. As used herein, a “processor” encompasses any one or functional combination of the foregoing examples.

Operational aspects disclosed herein may be embodied directly in hardware, in a software module executed by a processor, or in a combination of the two. A software module may reside in RAM memory, flash memory, ROM memory, EPROM memory, EEPROM memory, registers, hard disk, a removable disk, a CD-ROM, or any other form of storage medium known in the art. An exemplary storage medium is coupled to the processor such the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an ASIC. The ASIC may reside in a user terminal. In the alternative, the processor and the storage medium may reside as discrete components in a user terminal.

Furthermore, the one or more versions may be implemented as a method, apparatus, or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware, or any combination thereof to control a computer to implement the disclosed aspects. Non-transitory computer readable media can include but are not limited to magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips . . .), optical disks (e.g., compact disk (CD), digital versatile disk (DVD), BluRay™ . . .), smart cards, solid-state devices (SSDs), and flash memory devices (e.g., card, stick). Of course, those

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skilled in the art will recognize many modifications may be made to this configuration without departing from the scope of the disclosed aspects.

In view of the exemplary systems described supra, methodologies that may be implemented in accordance with the disclosed subject matter have been described with reference to several flow diagrams. While for purposes of simplicity of explanation, the methodologies are shown and described as a series of blocks, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methodologies described herein. Additionally, it should be further appreciated that the methodologies disclosed herein are capable of being stored on an article of manufacture to facilitate transporting and transferring such methodologies to computers.

The previous description of the disclosed aspects is provided to enable any person skilled in the art to make or use the present disclosure. Various modifications to these aspects will be clear to those skilled in the art, and the generic principles defined herein may be applied to other embodiments without departing from the spirit or scope of the disclosure. Thus, the present disclosure is not intended to be limited to the embodiments shown herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

The invention claimed is:

1. A method for synchronizing audio output by an external sound system driven by a host player device with video output from a head-mounted display (HMD) player device, the method comprising:

establishing a session between the host player device and the HMD player device, wherein the host player device is coupled to an external sound system and the HMD player device comprises a video output system;

determining, by at least one of the host player device or HMD player device, a latency period between audio output from the external sound system and corresponding video output from the HMD player device, wherein the determining is based on both the host and HMD player devices simultaneously initiating play of test content;

providing, by the host player device, both the host and HMD player devices with access to identical digital audio-video content selected by a user;

initiating play of the digital audio-video content by the host player device for audio output by the external sound system; and

initiating play of the digital audio-video content by the HMD player device after waiting for the latency period so that audio output by the external sound system is synchronized with video output from the HMD player device.

2. The method of claim 1, wherein the determining comprises determining the latency period based on user input received by the host player device in response to contemporaneous output of the test content from the host and HMD player devices.

3. The method of claim 1, wherein the determining comprises determining the latency period based on detecting, by at least one of the host player device or HMD player device, a lag between audio output from the external sound system and corresponding video output from the HMD player device, wherein both the host and HMD player devices simultaneously initiate play of the test content.

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4. The method of claim 3, wherein the host player device sends a wireless signal to the HMD player device to instruct initiation of the corresponding video output.

5. The method of claim 4, wherein the signal comprises at least one of an audio signal or an optical signal emitted by the external sound system.

6. The method of claim 1, wherein establishing a session further comprises verifying a connection between the first and second applications.

7. The method of claim 1, wherein establishing the session is performed over a Wi-Fi connection.

8. The method of claim 1, further comprising signaling by the HMD player device to the host player device that the HMD player device is ready to play content.

9. The method of claim 1, further comprising ensuring, by the host player device in cooperation with the HMD player device, that the host player device and the HMD player device contain identical copies of the digital audio-video content.

10. The method of claim 9, wherein the ensuring further comprises querying by the host player device whether the memory of the HMD player device contains a copy of the digital audio-video content.

11. The method of claim 9, wherein the ensuring further comprises copying the content to be played from the host player device to the HMD player device.

12. The method of claim 1, further comprising causing deletion after play, by at least one of host player device or the HMD player device, of the digital audio-video content on the HMD player device based on satisfaction of a predetermined condition.

13. The method of claim 12, wherein the predetermined condition comprises at least one of: number of plays, a time-period, or a license type.

14. The method of claim 1, further comprising displaying by the host player device at least one of: a navigation screen with playback controls and latency compensation, video output of the digital audio-video content, overlaid navigation controls, or a 3D video output version of the digital audio-video content.

15. The method of claim 1, wherein the external sound system comprises a home theater system.

16. An apparatus for synchronizing audio output by an external sound system driven by a host player device with video output from a head-mounted display (HMD) player device, comprising at least one processor coupled to a memory, the memory holding program instructions, that when executed by the at least one processor, cause the apparatus to perform:

establishing a session between the host player device and the HMD player device;

determining a latency period between audio output from the external sound system and corresponding video output from the HMD player device;

providing both the host and HMD player devices with access to identical digital audio-video content selected by a user;

initiating play of the digital audio-video content for audio output by the external sound system; and

causing play of the digital audio-video content to be initiated by the HMD player device after waiting for the latency period so that audio output by the external sound system driven is synchronized with video output from the HMD player device.

17. The apparatus of claim 16, wherein the memory holds further instructions for the determining at at least in part by determining the latency period based on user input received

in response to contemporaneous output of the test content from the host and HMD player devices.

18. The apparatus of claim **16**, wherein the memory holds further instructions for the determining the latency period at least in part by detecting a lag between audio output from the external sound system and corresponding video output from the HMD player device, wherein both the host and HMD player devices simultaneously initiate play of the test content.

19. The apparatus of claim **16**, wherein the memory holds further instructions for sending a signal to the HMD player device to instruct initiation of the corresponding video output.

20. The apparatus of claim **16**, wherein the memory holds further instructions for the determining based on both the host and HMD player devices simultaneously initiating play of test content.

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